Purpose of Each Task in FreeRTOS

Project: Multi-Tasking RTOS Firmware Scheduler

This document explains the purpose of the three independent tasks specified in the assessment: Sensor Acquisition, Display Updates, and Emergency Alarm Handling. It also describes task priorities and the synchronization mechanisms used to coordinate them.

1. Sensor Acquisition Task (SensorTask)

Purpose: Read measurements from the connected sensor at defined intervals and make the latest readings available to the other tasks.

Main responsibilities

• Read the sensor periodically, for example every 5 seconds in a demonstration.

• Check whether the sensor reading is valid.

• Send temperature and humidity values to a queue so other tasks can use the same data.

• Avoid delaying time-critical alarm handling with long blocking operations.

Why it is needed: The display and alarm tasks require current sensor data. Keeping acquisition in its own task makes sampling periodic and independent of display refreshes.

2. Display Updates Task (DisplayTask)

Purpose: Display the latest sensor measurements on the local LCD at regular intervals.

Main responsibilities

• Receive the latest sensor values from the queue.

• Update the LCD with readable temperature and humidity values.

• Use a mutex or another suitable protection mechanism if multiple tasks can access the display.

• Keep display operations separate from sensor sampling and emergency monitoring.

Why it is needed: The LCD can be updated without making sensor acquisition wait for display operations. This separation improves modularity and helps avoid shared-resource conflicts.

3. Emergency Alarm Handling Task (AlarmTask)

Purpose: Monitor sensor values for predefined critical conditions and activate the alarm output promptly when a threshold is exceeded.

Main responsibilities

• Check the latest available measurement against the configured threshold.

• Activate the buzzer and warning LED when a critical condition is detected.

• Deactivate the alarm when the reading returns to the defined safe range, if that behavior is required.

• Use a high task priority or task notification where appropriate so response is not unnecessarily delayed.

Why it is needed: Emergency handling is time-sensitive. An independent, appropriately prioritized task can respond without waiting for routine LCD updates.

4. Example Task Priority and Execution

Task	Example priority	Execution	Purpose
AlarmTask	4 (highest)	Monitors for critical readings / responds to notification	Prompt emergency response
SensorTask	3	Periodic sensor sampling	Provide fresh measurements

Task	Example priority	Execution	Purpose
DisplayTask	2	Periodic LCD refresh	Show current readings

5. Synchronization and Communication

• **Queue:** Transfers sensor measurements from SensorTask to tasks that need the values.

• **Mutex:** Protects a shared LCD or other shared resource from simultaneous access.

• **Task notification or semaphore:** Can signal AlarmTask when a critical reading is detected, reducing unnecessary waiting.

6. Expected Result

During normal operation, SensorTask periodically acquires measurements and DisplayTask updates the LCD. When a reading exceeds the configured critical threshold, AlarmTask should activate the buzzer and LED promptly. The tasks communicate through suitable RTOS mechanisms and should not interfere with one another unnecessarily. Priority values and timing intervals in this document are examples; configure them to match the actual firmware and hardware used.